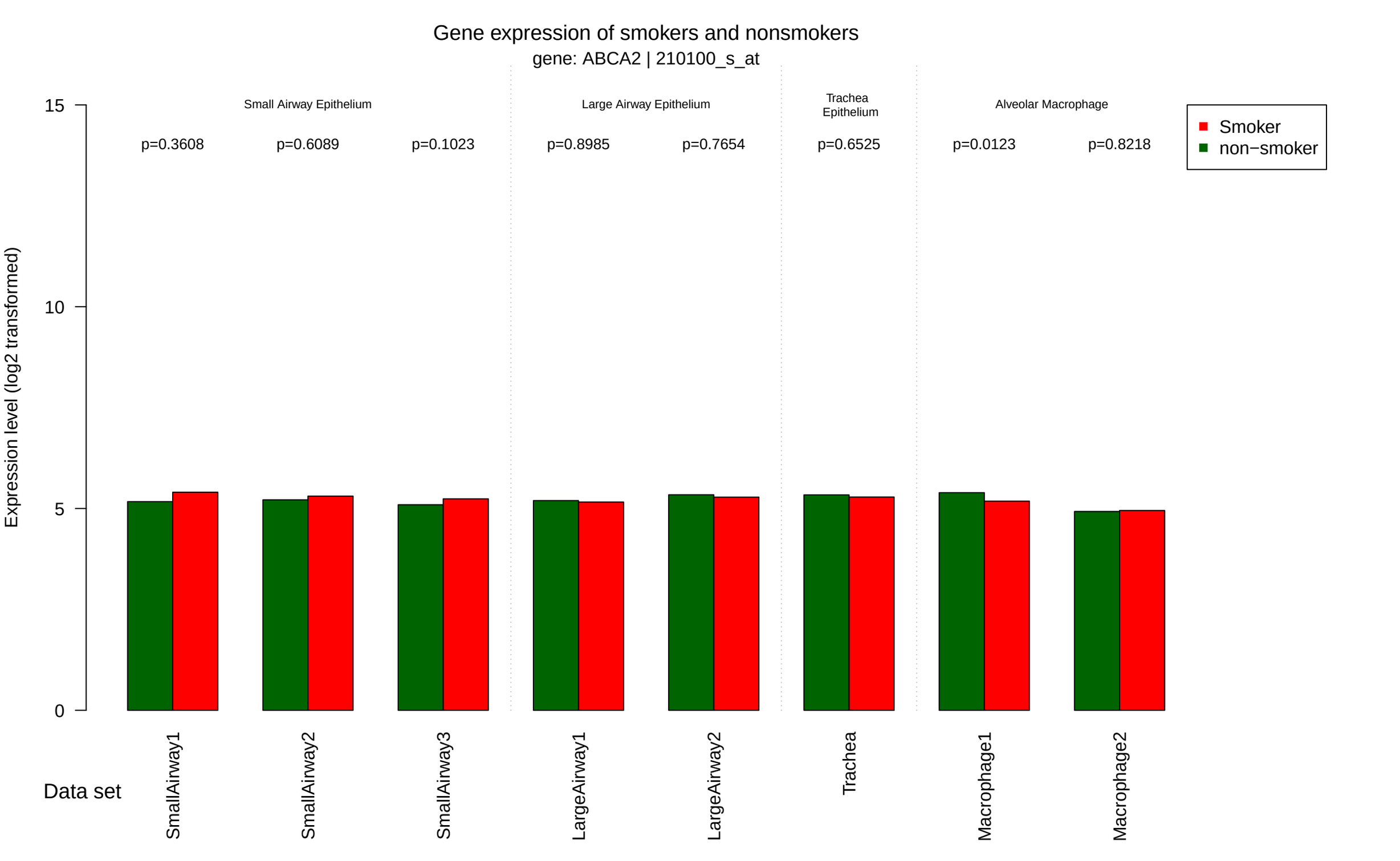
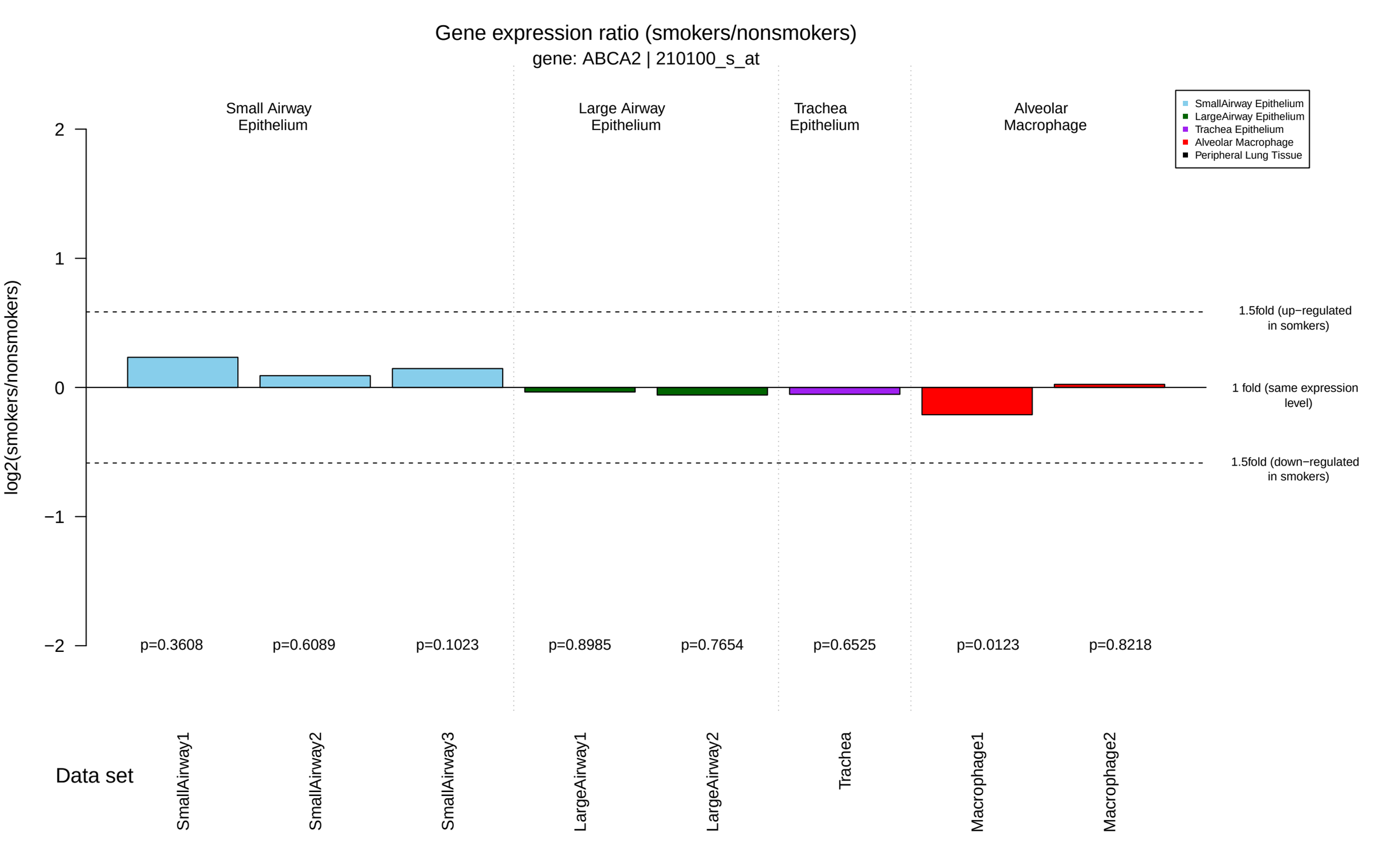
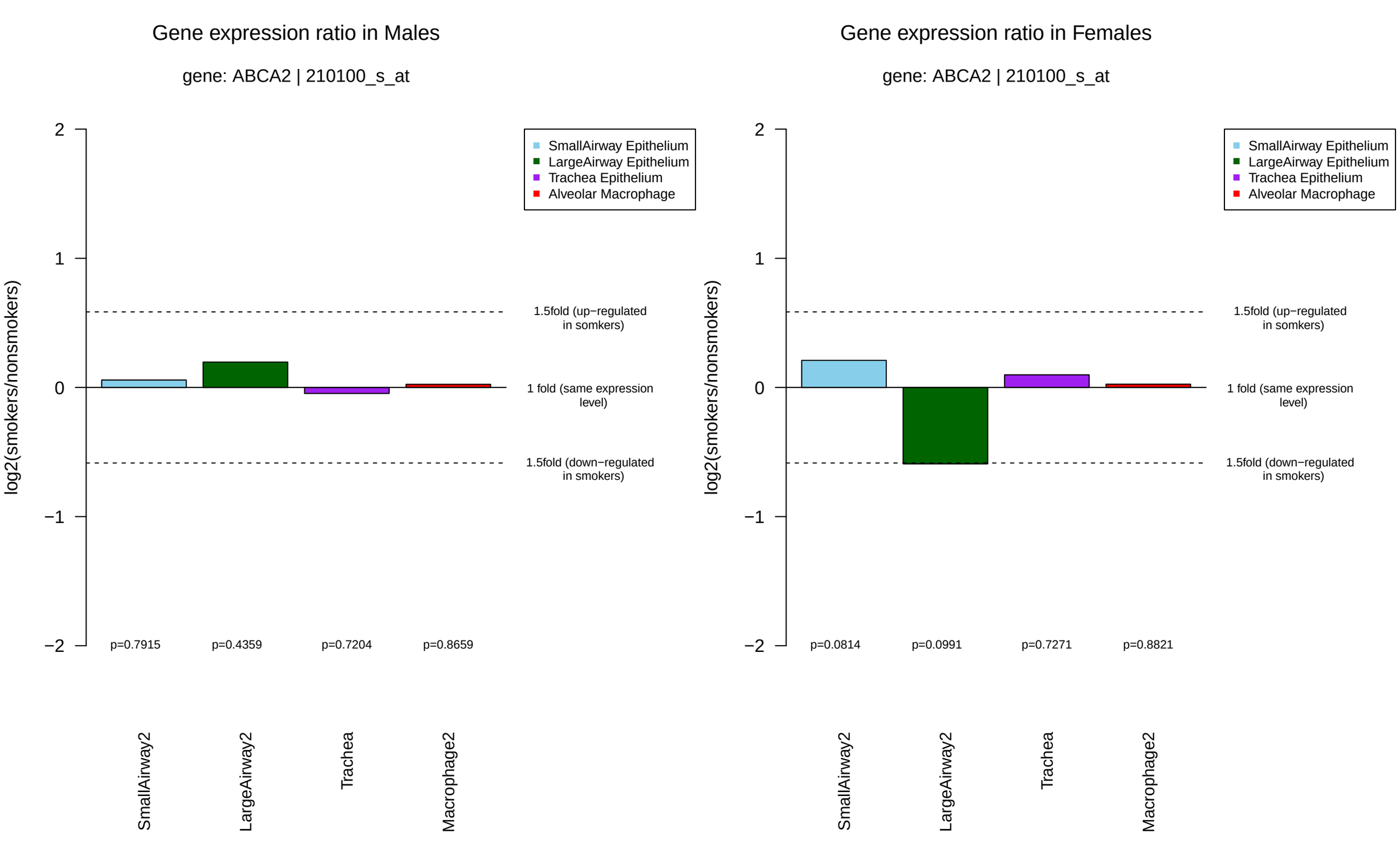
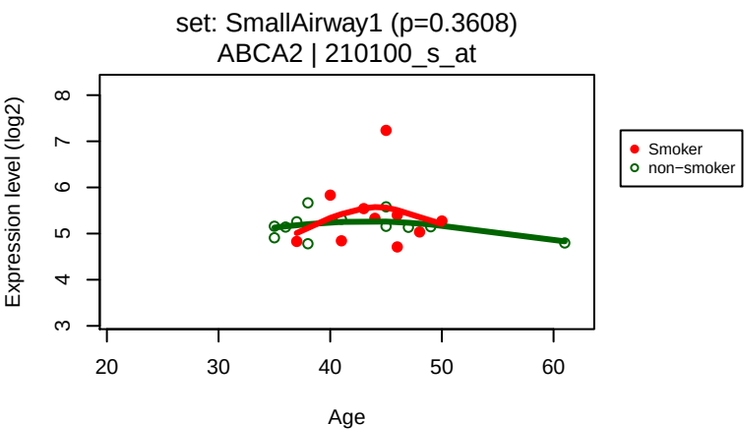
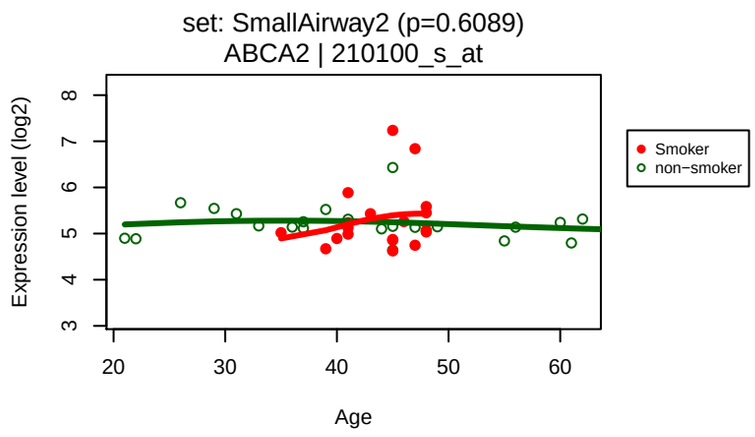
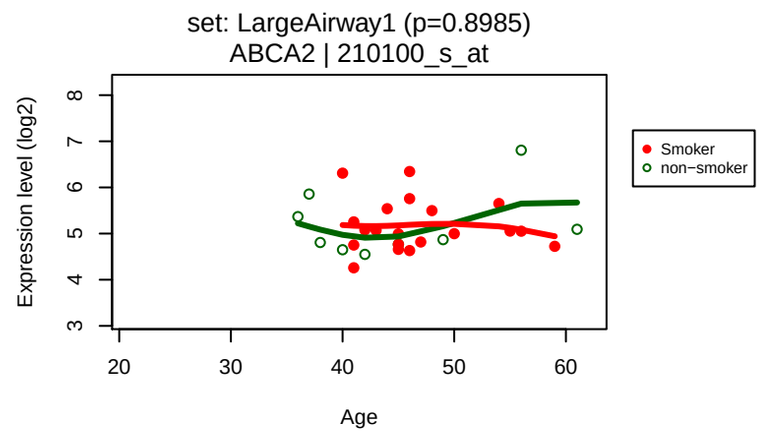

Gene expression of smokers and nonsmokers
gene: ABCA2 | 210100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3608
p=0.6089
p=0.1023
p=0.8985
p=0.7654
p=0.6525
p=0.0123
p=0.8218
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

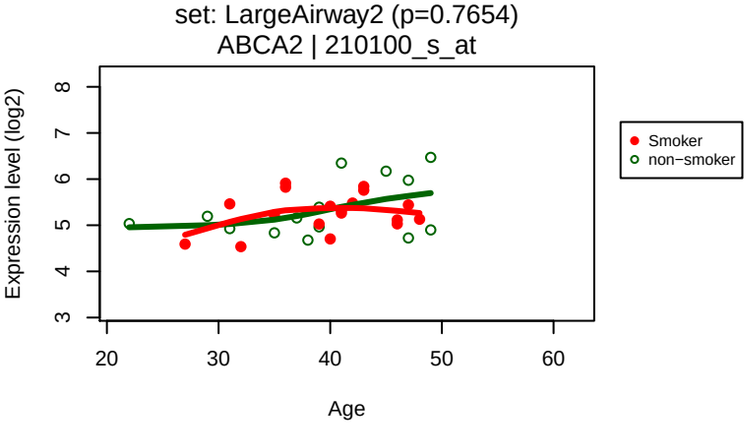

Gene expression ratio (smokers/nonsmokers)
gene: ABCA2 | 210100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3608
p=0.6089
p=0.1023
p=0.8985
p=0.7654
p=0.6525
p=0.0123
p=0.8218
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

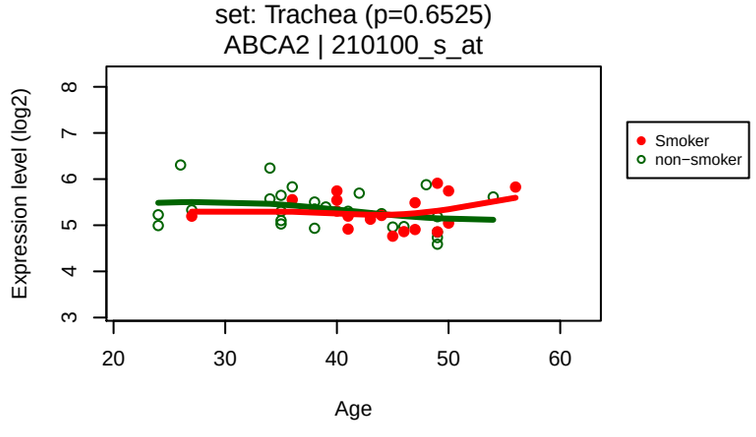

Gene expression ratio in Males
gene: ABCA2 | 210100_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7915
p=0.4359
p=0.7204
p=0.8659
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCA2 | 210100_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0814
p=0.0991
p=0.7271
p=0.8821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

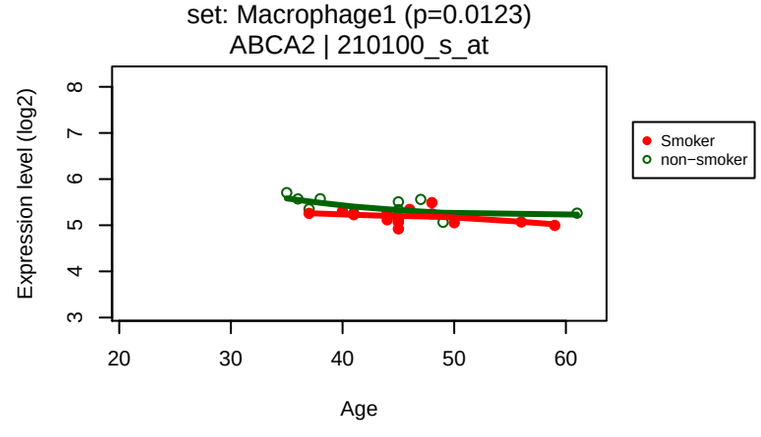

set: SmallAirway1 (p=0.3608)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker

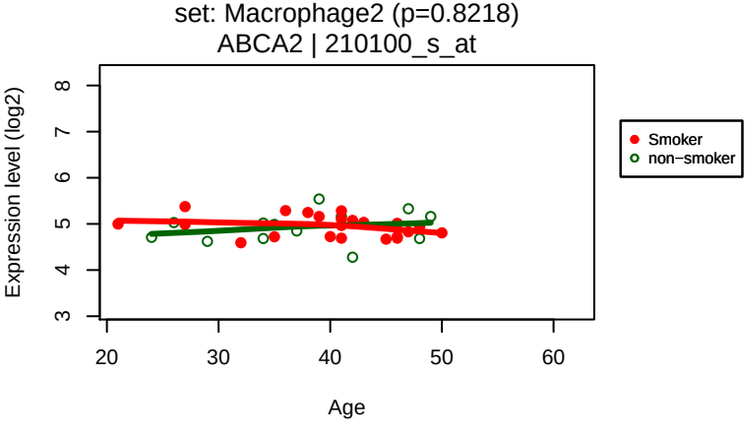

set: SmallAirway2 (p=0.6089)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker

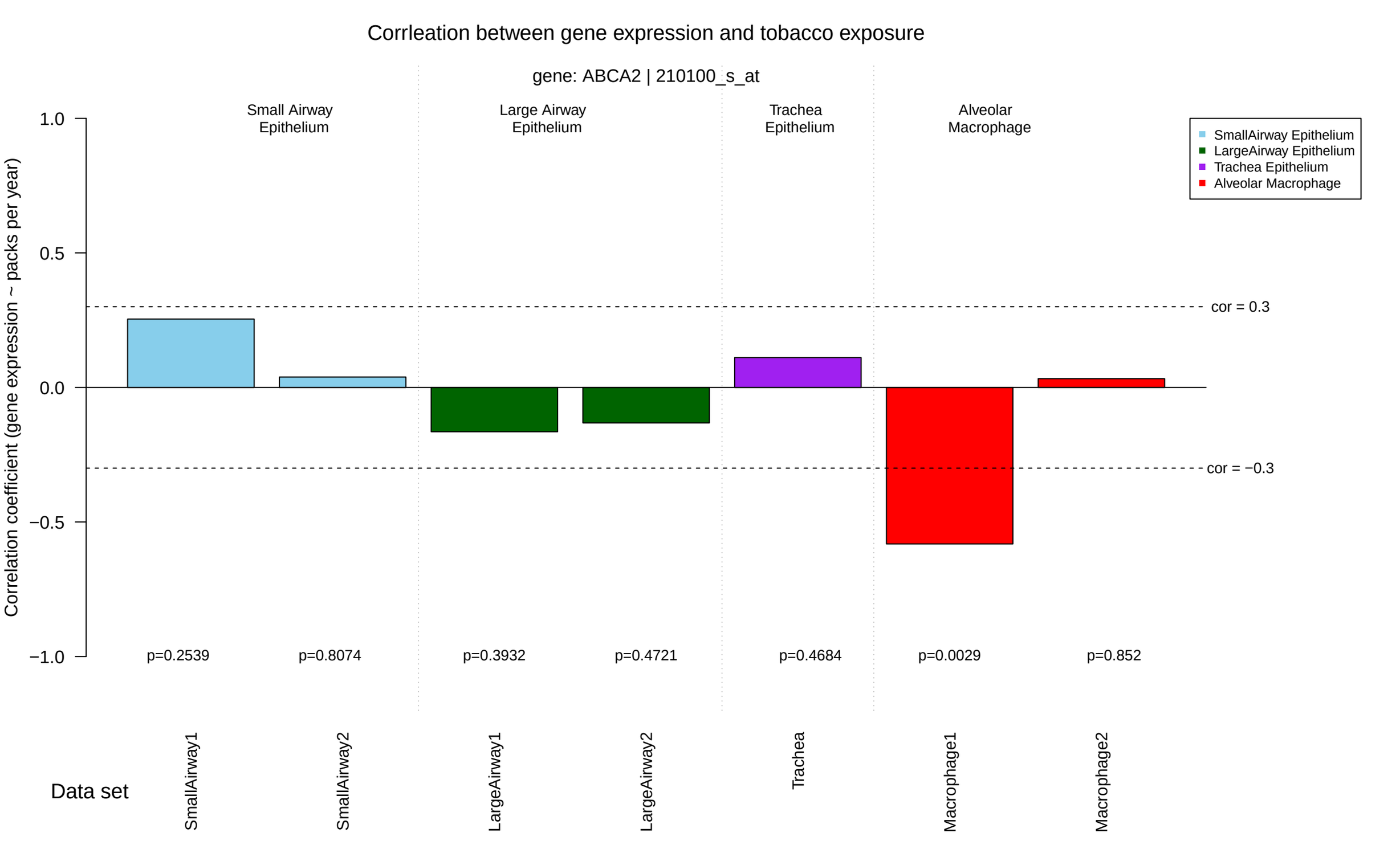

set: LargeAirway1 (p=0.8985)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7654)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6525)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0123)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8218)
ABCA2 | 210100_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCA2 | 210100_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2539
p=0.8074
p=0.3932
p=0.4721
p=0.4684
p=0.0029
p=0.852
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2